

Mini Lesson Sample Script

Materials

(S) Personal white board

PROBLEM	SAMPLE SCRIPT
1 Model a multiplication word problem involving units of 6 by drawing and labeling a tape diagram, then solve by breaking the factor into two smaller facts. Rosa has 7 boxes of crayons. Each box has 6 crayons. How many crayons does Rosa have in all? Draw and label a tape diagram to model the problem. Write an equation using a letter for the unknown. Solve. <i>Answer: $c = 42$ crayons; $7 \times 6 = c$; $(5 \times 6) + (2 \times 6) = 30 + 12 = 42$</i>	T: Read this problem. What do we know? S: There are 7 boxes and each box has 6 crayons. T: What are we trying to find? S: How many crayons there are in all. T: We can draw a tape diagram to model this. How many equal parts should we draw? S: 7, because there are 7 boxes. T: Right. Draw 7 equal parts on your board. What goes inside each part? S: 6, because each box has 6 crayons. T: Label each part "6." Now, the total is what we're trying to find. Let's use the letter c for the total number of crayons. Write "c crayons" above the whole tape. S: (Label the tape diagram.) T: Write an equation for this problem. S: $7 \times 6 = c$. T: Do you know 7×6 right away? Here's a way to break it into facts you might already know. What if we split the 7 groups into 5 groups and 2 groups? (Draw a line on the tape diagram between the 5th and 6th parts.) What is 5×6? S: 30. T: And 2×6? S: 12. T: So what is $30 + 12$? S: 42. T: So $c = 42$ crayons. Write a sentence to answer the problem. S: Rosa has 42 crayons in all.
2 Model a division word problem involving units of 6 by drawing a tape diagram with an unknown number of groups, then solve using a related multiplication strategy or by decomposing the dividend. Marcus has 54 stickers. He puts 6 stickers on each page. How many pages does Marcus use? Draw and label a tape diagram to model the problem. Write an equation using a letter for the unknown. Solve. <i>Answer: $p = 9$ pages; $p \times 6 = 54$ or $54 \div 6 = p$; $(30 \div 6) + (24 \div 6) = 5 + 4 = 9$</i>	S: There are 54 stickers total, and each page gets 6 stickers. T: What are we trying to find? S: How many pages Marcus uses. T: So this time, do we know the total or are we looking for it? S: We know the total. We don't know how many groups. T: Good. Let's use the letter p for the number of pages. On your tape diagram, label the top "54 stickers." Each part gets "6 stickers." We don't know yet how many parts there are — that's p. Write an equation. S: $54 \div 6 = p. \rightarrow p \times 6 = 54$. T: Now, can we break 54 into parts that are easier to divide by 6? What's a number you know that divides evenly by 6? S: 30. $30 \div 6 = 5$. T: If we've used 30 of the 54 stickers, how many are left? S: 24. T: What's $24 \div 6$? S: 4. T: So how many pages is that altogether? S: $5 + 4 = 9$ pages. T: $p = 9$ pages. Write a sentence to answer the problem. S: Marcus uses 9 pages.
3 Solve a word problem involving units of 7 by writing an equation with a letter for the unknown, drawing a model, and choosing a strategy to find the answer. A parking lot has 6 rows of cars. Each row has 7 cars. How many cars are in the parking lot? Write an equation using a letter for the unknown. Draw a tape diagram and solve. <i>Answer: $t = 42$ cars; $6 \times 7 = t$; $(5 \times 7) + (1 \times 7) = 35 + 7 = 42$</i>	T: Now try this one on your own. Read the problem, write an equation with a letter for the unknown, draw a tape diagram, and solve. S: (Work.)